

Null-space energy placement accompanies lower activation range and quantization error. All 28 Llama-3.2-3B down-input layers are shown for Hadamard, DuQuant and PrismQuant at maximal alignment rank. (a) Whole-activation energy fraction in the groupwise quantizer null space; the dashed reference is $1/128$. (b) Paired mean group range. (c) Dynamic asymmetric group-128 INT4 activation NMSE. The arithmetic means of per-layer PrismQuant-versus-Hadamard percentage reductions are 25.30% and 40.41%, respectively. These are descriptive layer measurements, without inferred seed confidence intervals. DuQuant denotes the existing DuQuant-style diagnostic, not the complete official implementation. The shared legend applies to all panels.